RGB MOOD LAMP

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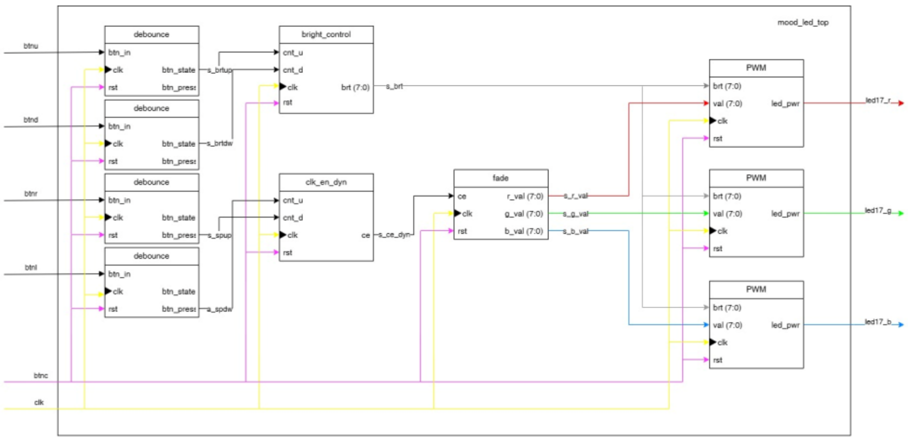
PROBLEM DESCRIPTION

VHDL code for an RGB lamp to enlighten your day. The project features fading colours in a colour wheel loop with adjustable brightness and speed of colour change. Designed specifically for the Nexys A7-50T board utilizing the onboard RGB LED and 5 buttons.

BLOCK DIAGRAM



Hierarchy of VHDL modules and signal flows between toplevel and individual components.

MODULE: BRIGHT CONTROL & PWM

Bright Control:

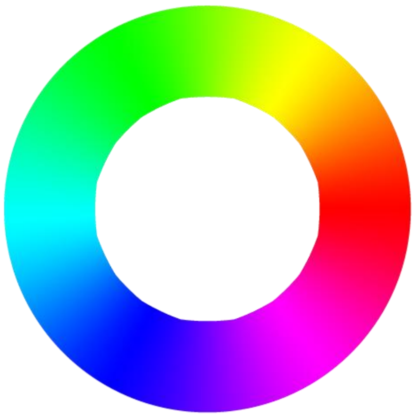
This block manages brightness using an up-down counter with an integrated auto-repeat function. The resulting 8-bit vector serves as a global intensity control for all PWM blocks.

PWM:

This module implements a PWM generator with an integrated hardware multiplier. It calculates the duty cycle by multiplying two 8-bit inputs: global brightness and colour value. This is achieved by latching the calculated duty cycle into a shadow register once the counter reaches its maximum value.

COLOUR WHEEL CHANGE

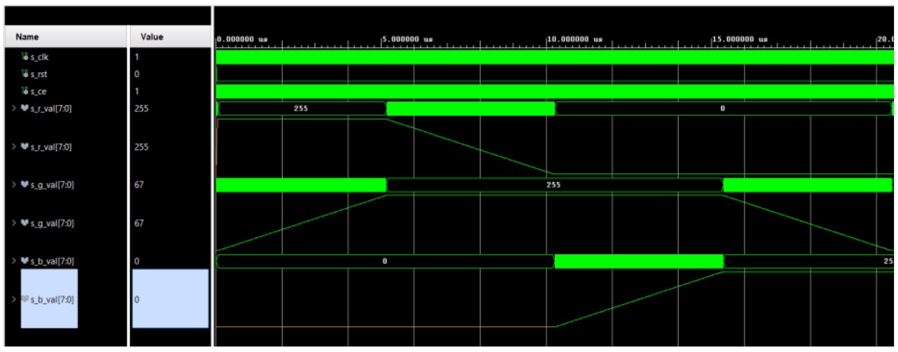
RGB LEDs consist of 3 singlecoloured LEDs. To ensure consistent fade of colours we're using colour wheel principle. First LED is at it's maximum brightness, second on it's minimum and third changes it's brightness.



MODULE: FADE & CLK_EN_DYN

Fade:

A block utilizing a fading algorithm where colours transition smoothly around the colour wheel. There are six states in total. An internal counter ensures the correct sequence of colours. Outputs are 3 unique 8-bit vectors - one for each colour.



Clk_en_dyn:

Modified version of clk_en using an up-down counter, which allows us to control how often the signal is sent to the Fade block, effectively changing the speed of colour fading. There are 9 speed modes in total. The output frequency ranges from 111 Hz to 1000 Hz with default frequency of 200 Hz.

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Team members:

Ondřej Kovář
- ✓ README, Bright control, PWM, block diagram

Richard Královský
- ✓ Simulations, clk_en_dyn, fade, top level, implementation